

minilab3/minilab3-dr.c

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <pthread.h>
4
5  int sharedCounter = 0;
6  int num_iterations;
7
8  void *thread(void *vargp)
9  {
10     for (int i = 0; i < num_iterations; i++) {
11         sharedCounter++;
12     }
13 }
14
15 int main(int argc, char *argv[]) {
16     if (argc != 3) {
17         printf("Usage: %s <#_iteration> <#_thread>\n", argv[0]);
18         exit(1);
19     }
20
21     num_iterations = atoi(argv[1]);
22     int num_threads = atoi(argv[2]);
23
24     if (num_threads > 10) {
25         printf("Maximal number of threads is 10\n");
26         exit(1);
27     }
28
29     pthread_t tids[num_threads];
30
31     for (int i = 0; i < num_threads; i++) {
32         pthread_create(&tids[i], NULL, thread, NULL);
33     }
34     for (int i = 0; i < num_threads; i++) {
35         pthread_join(tids[i], NULL);
36     }
37
38     printf("Threads: %d, Expected Iterations: %d, Actual Iterations: %d\n",
39         num_threads, num_iterations * num_threads, sharedCounter);
```

```
40 |     exit(0);  
41 | }  
42 |
```